

Big Data Systems & Statistics Project

Daragh Brady

D00255640

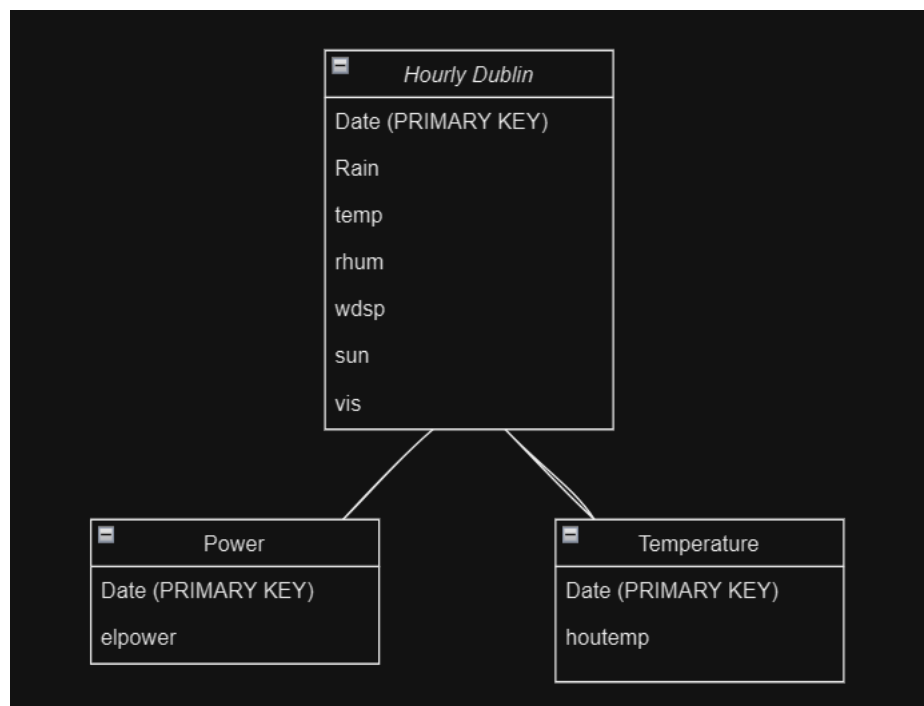
Project Description

For this project I had to clean and analyse data from different csv files. My goal was to show that the Electrical Power could be predicted based on different factors. I had to understand how to merge csv files based on the date (Zach, 2022) so I could perform the analysis.

Research Methodology

I loaded the csv files into a Mongo DB database. I chose this because I wanted to perform analysis where I could use aggregate queries that I cannot do in SQL, and the data is not very structured in the original files, as the dates are not matching. I done this with my hourly_dublin_17_18_19.csv, JLHome1718Power.csv, and JLHome1718Temperature.csv files. However, after I completed cleaning the dataset in R, I chose to use SQL to analyse the dataset even further, as the data was a lot more structured so I could use the date column properly as I had organised it better and the data was all beginning from the 12th of October 2017. For handling different data from homes and weather stations, with MongoDB's ability to store flexible documents, it can modify fields or create new ones. For scaling every home in Britain and Ireland, MongoDB allows auto-scaling measures (Shivangi, 2023). For backing up the database, I created multiple replicas sets in MongoDB, and had the SQL database stored in a XAMPP environment. For database security, I added a secondary user for both the SQL server, and the MongoDB database (Tutorials Teacher, 2023).

This is my schematic of my database using a class diagram. Created on draw.io.



Data Wrangling and Cleaning

I had to manually edit the excel files so I could load them into my software so I could clean and prepare the data. I done this my adding a row to the JLHome1718Temperature.csv file and calling the first column “date” and the second “houtemp” so I could distinguish it from the air temperature in the hourly_dublin_17_18_19.csv file when they are joined. I also added the date and elpower columns to the JLHome1718Power.csv file. This way I could load in the files and match them based on the date value. I then merged the files based on the date, so each date had a value, however I encountered my first issue when I realised the hourly_dublin_17_18_19.csv file date column started on the 1st of January 2017, and the other files started on the 12th of October 2017.

```
'data.frame': 1017131 obs. of 9 variables:
 $ date : chr "01/01/2017 00:00" "01/01/2017 01:00" "01/01/2017 02:00" "01/01/2017 03:00" ...
 $ rain : num 0.9 0.2 0.1 0 0 0 0 0 0 ...
 $ temp : num 5.3 4.9 5 4.2 3.6 2.8 1.7 1.6 2 2.6 ...
 $ rhum : int 91 95 92 90 88 89 91 91 89 84 ...
 $ wdsp : int 12 8 8 12 11 12 13 13 13 13 ...
 $ sun : num 0 0 0 0 0 0 0 0 0 0 ...
 $ vis : int 25000 25000 30000 30000 30000 30000 30000 30000 35000 35000 35000 ...
 $ elpower: int NA NA NA NA NA NA NA NA NA NA ...
 $ houtemp: num NA NA NA NA NA NA NA NA NA NA ...
```

To resolve this, I removed the rows where a NA value was. This meant I had the Precipitation Amount, Air Temperature, Relative Humidity, Mean Hourly Wind Speed, Sunshine Duration, Visibility, Electrical Power, and House Temperature values from the 12th of October 2017.

```
'data.frame': 16690 obs. of 9 variables:
 $ date : chr "12/10/2017 23:00" "13/10/2017 00:00" "13/10/2017 01:00" "13/10/2017 02:00" ...
 $ rain : num 0 0 0 0 0 0 0 0 0 0 ...
 $ temp : num 16.5 16.5 17 16.9 16.7 16.8 17 17.2 17.1 17.4 ...
 $ rhum : int 86 84 86 86 87 87 87 86 88 86 ...
 $ wdsp : int 17 15 15 15 13 11 18 20 21 20 ...
 $ sun : num 0 0 0 0 0 0 0 0 0 0.2 ...
 $ vis : int 30000 30000 30000 30000 30000 30000 30000 30000 30000 40000 ...
 $ elpower: int 39 26 24 26 271 134 27 131 165 123 ...
 $ houtemp: num 16.3 16.3 16.3 16.3 16.3 16.3 16.3 16.3 16.4 16.9 16.6 ...
```

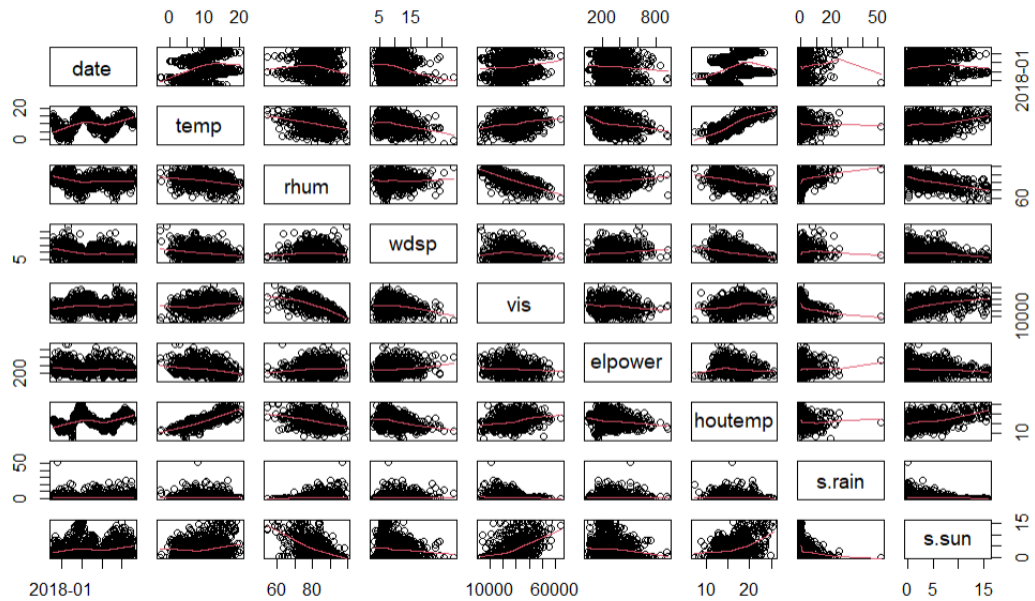
Next, I changed the date column to show each day, and made the columns have averages of that day rather than per hour. I realised that the Sunshine Duration and Precipitation Amount should be totals for that day rather than an average of the day. After this I was happy with the dataset as it was able to be analysed.

```
tibble [724 × 9] (S3: tbl_df/tbl/data.frame)
 $ date : Date[1:724], format: "2017-10-12" "2017-10-13" ...
 $ temp : num [1:724] 16.5 16.5 15.2 14.3 13.9 ...
 $ rhum : num [1:724] 86 88.7 86.6 86.7 84.4 ...
 $ wdsp : num [1:724] 17 14.12 9.67 10.92 20.5 ...
 $ vis : num [1:724] 30000 23333 20750 20304 12767 ...
 $ elpower: num [1:724] 39 301 307 232 493 ...
 $ houtemp: num [1:724] 16.3 16.8 17.2 17.2 17.6 ...
 $ s.rain : num [1:724] 0 3.9 3.5 1.7 0.8 0 0.1 6.1 3.1 6.2 ...
 $ s.sun : num [1:724] 0 1.4 1.7 0.7 1 7.2 1.3 0 2 0.9 ...
```

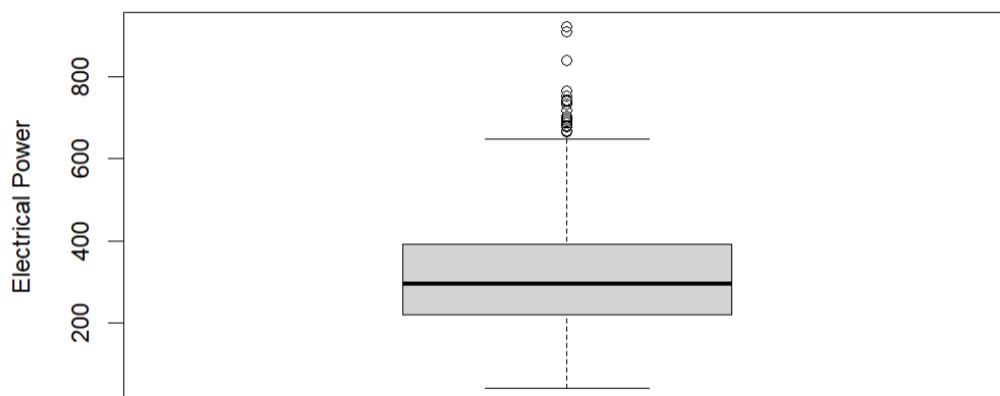
Exploratory Analysis

First, I loaded in the csv files to R, and through different types of data cleaning, I was able to start my analysis of the data set.

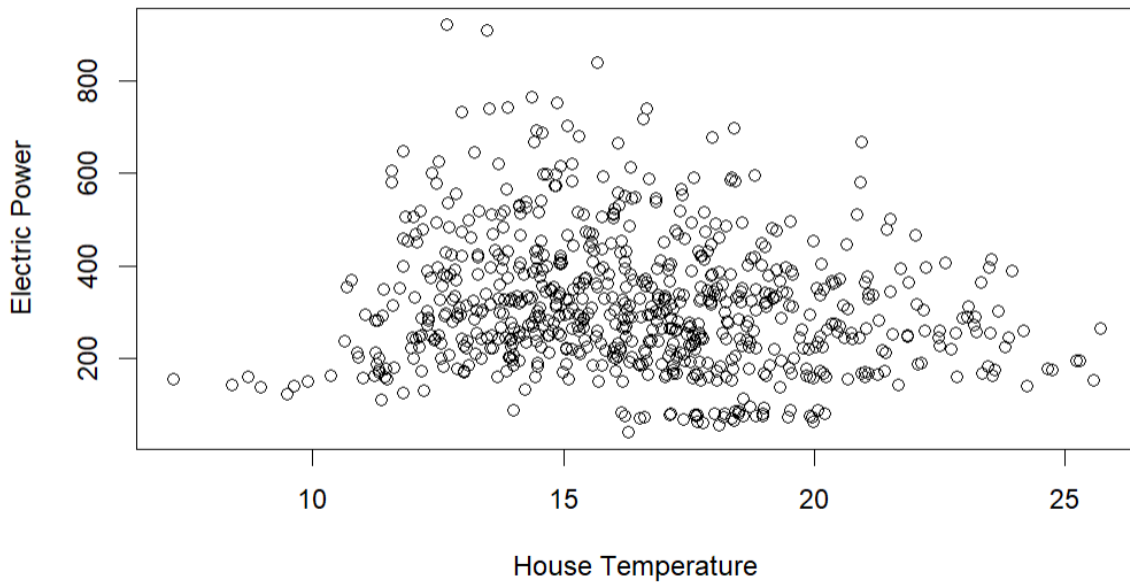
Next, I used a pairs plot of the columns to try and get a simple understanding of what possible columns could have a relationship.



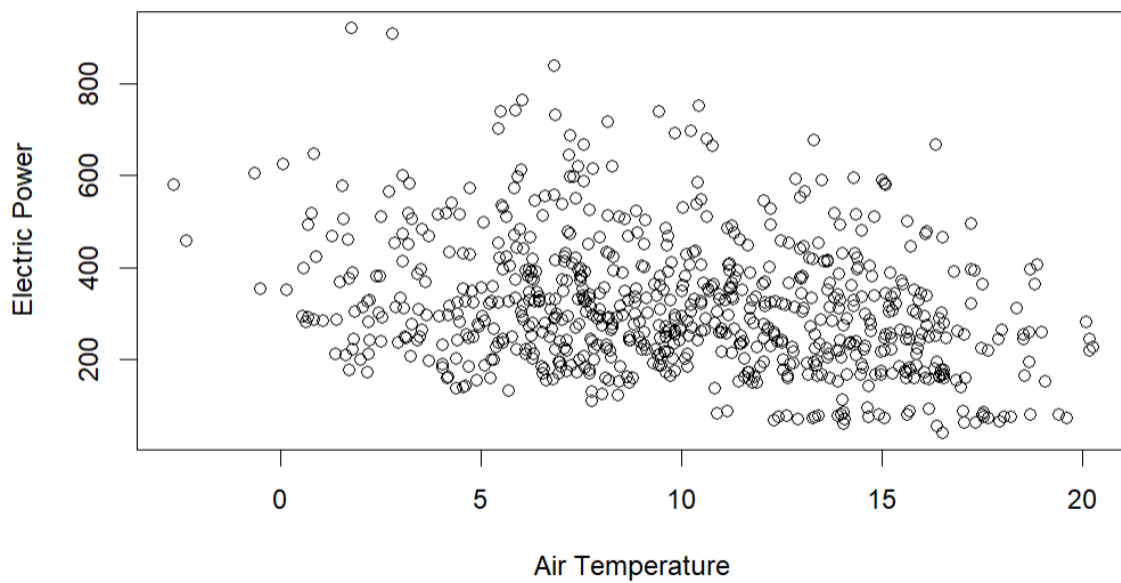
Upon deciding that Electrical Power was the response variable, I decided to explore it through different visualisations and finding correlations.



This was a boxplot to show the distribution of the Electrical Power column. As you can see there are quite a few outliers, I decided to keep them in the data set as there would be some days throughout the data where there were cases of when it would be very necessary to use lots of electrical power, especially if it is a big house.



This was a scatterplot to show the distribution of Electrical Power with the House Temperature. As you can see, the distribution shows that the electrical power is quite spread throughout the range of the house temperature values.



This was a scatterplot to show the distribution of Electrical Power with the Air Temperature. Also, in this graph you can see there is not much of a linear relationship between the Air Temperature and Electrical Power values.

date	temp	rhum	wdsp	vis
Min. :2017-10-12	Min. :-2.683	Min. :54.04	Min. : 3.167	Min. : 2804
1st Qu.:2018-04-10	1st Qu.: 6.534	1st Qu.:76.76	1st Qu.: 6.832	1st Qu.:21646
Median :2018-10-08	Median : 9.760	Median :81.79	Median : 9.134	Median :27688
Mean :2018-10-09	Mean : 9.900	Mean :81.55	Mean : 9.861	Mean :28056
3rd Qu.:2019-04-07	3rd Qu.:13.677	3rd Qu.:86.53	3rd Qu.:11.917	3rd Qu.:34198
Max. :2019-10-13	Max. :20.254	Max. :99.50	Max. :28.750	Max. :63750

elpower	houtemp	s.rain	s.sun
Min. : 39.0	Min. : 7.213	Min. : 0.000	Min. : 0.000
1st Qu.:219.1	1st Qu.:14.128	1st Qu.: 0.000	1st Qu.: 0.500
Median :294.2	Median :16.248	Median : 0.200	Median : 2.800
Mean :315.1	Mean :16.450	Mean : 1.987	Mean : 3.921
3rd Qu.:391.7	3rd Qu.:18.422	3rd Qu.: 2.000	3rd Qu.: 6.225
Max. :922.9	Max. :25.704	Max. :51.900	Max. :15.900

After running the summary command on my dataset, I got a good understanding of each column. I found that the data in general is quite spread, with potential outliers in a lot of the columns. This was the data after I cleaned the merged file so I could accurately see the data that I was going to perform my analysis on.

I also explored the data with MongoDB. I was able to load in the different files to my database, and I had each file in an individual collection. Through this I was able to perform some simple aggregation queries to find out some useful information about the dataset.

```
< {
  _id: '2017-01-11',
  avg_power: 225.19722222222222
}
{
  _id: '2017-01-12',
  avg_power: 273.47430555555553
}
{
  _id: '2017-02-11',
  avg_power: 298.35555555555555
}
```

This code shows the average power used on each date in the dataset. It orders the date on the first in the dataset. This provides some analysis where I could easily further explore the average power used in every month or year.

```
< {
  _id: '2017-01-11',
  avg_temperature: 15.826041666666665
}
{
  _id: '2017-01-12',
  avg_temperature: 12.020833333333334
}
{
  _id: '2017-02-11',
  avg_temperature: 15.714583333333334
}
```

This code shows the average house temperature on each date in the dataset. I could easily further explore the average temperature of each month or year.

```
< {
  _id: '2017-01-01',
  avg_temperature: 4.004166666666666
}
{
  _id: '2017-01-02',
  avg_temperature: 6.783333333333334
}
{
  _id: '2017-01-03',
  avg_temperature: 4.8374999999999995
}
```

This code shows the average air temperature on each date in the dataset. Also, I can explore the average air temperature of each month or year for comparison. The code for these aggregations can be found in the file with my code attached.

This is some of the analysis I was able to perform for this dataset through different SQL queries. This was performed with the combined clean dataset that I created through R.

date	elpower	wdsp
2018-06-25	364.666666666667	7.25
2018-06-27	194.833333333333	6.5416666666667
2018-06-28	152.5	6.45833333333333
2018-06-29	263.25	7.1666666666667
2018-07-04	243.458333333333	6.45833333333333
2018-07-06	160.25	5.25
2018-07-07	311.666666666667	6.58333333333333
2018-07-08	255.208333333333	5.375
2018-07-14	164.583333333333	6.08333333333333

This was the result I got when I found the date, electrical power, and wind speed of the days when the air temperature was above 17 degrees Celsius. This returned all of the selected columns for this filter. This is a small sample of the entire output.

Month ▲ 1	AvgHouseTemp	AvgElePower
2017-10	16.59626070487484	308.7428524374177
2017-11	14.384836956521736	343.3388888888889
2017-12	13.286438481423428	361.1799975249567
2018-01	12.956483362456348	344.4155570327431
2018-02	12.95351731601731	392.1406279408996
2018-03	13.48470046082949	365.56916282642084
2018-04	15.093913043478258	317.2184178743961
2018-05	19.20604838709677	303.9537634408602
2018-06	21.45389917027417	263.8697546897547
2018-07	21.642244373205102	245.4054882121151
2018-08	18.191666666666666	167.9731182795699
2018-09	17.369692152232005	296.07275385733976
2018-10	16.896992460525965	321.3063354595547

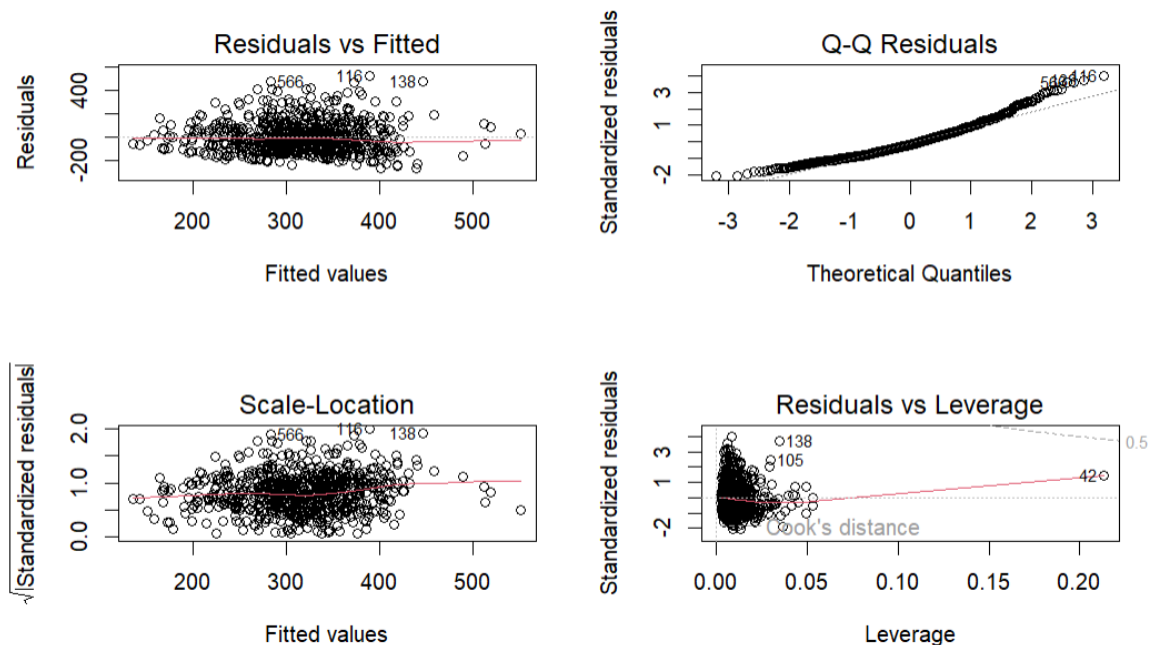
This was the output from the query where I found the average house temperature and electrical power per month of the dataset, this shows that in general, in the Winter, when houses are colder, people use more electrical power potentially to heat their house.

Month ▲ 1	AvgAirTemp
2017-10	11.741480566534916
2017-11	6.62025362318841
2017-12	5.418863611363201
2018-01	5.386091084867793
2018-02	3.4604025503482028
2018-03	4.476355606758832
2018-04	8.213381642512077
2018-05	11.986344086021509
2018-06	15.064495129870128
2018-07	16.535680057436718
2018-08	15.482795698924738
2018-09	12.316177566929118
2018-10	9.603954189264105

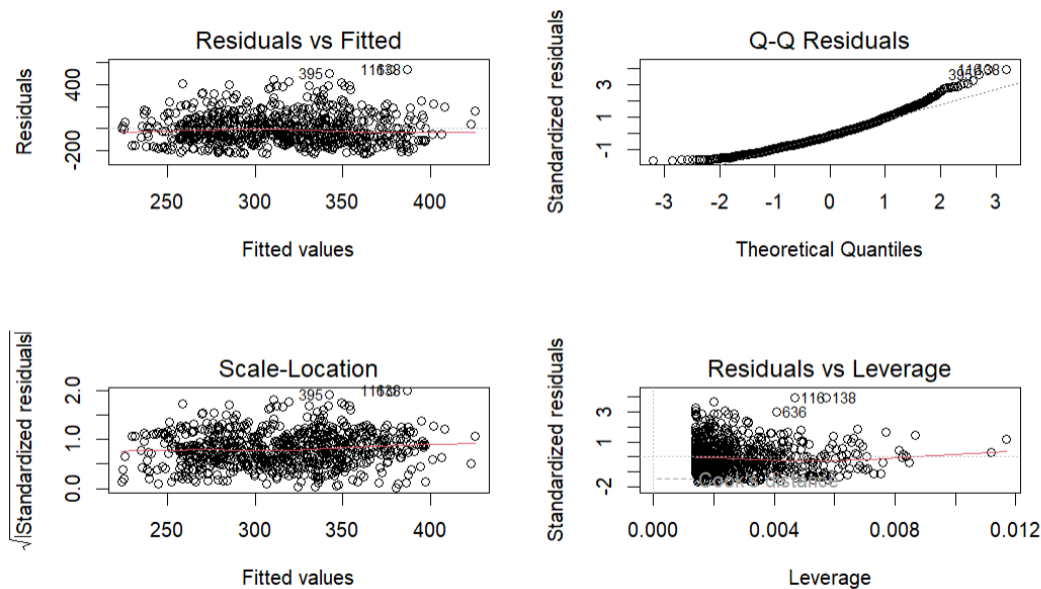
In this output I show the average air temperature of each individual month in the dataset. This shows me the difference in each month and I'm able to get a better understanding of the dataset.

Research Analysis and Findings

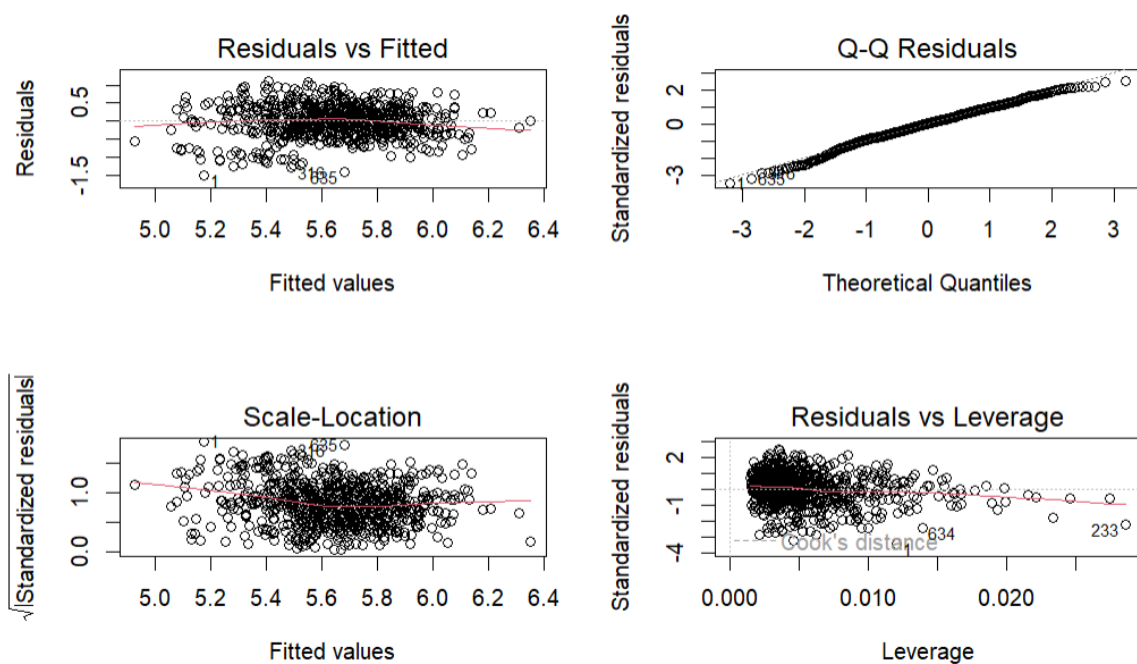
First, I calculated the linear model to predict Electrical Power with all the factors so I could understand what variables were the most significant to predict Electrical Power. These were the graphs I found through doing this.



I chose air temperature to be the factor to do my Simple Linear Regression as I found it best to predict the Electrical Power value. After doing this, I found that the Electrical Power value reduces by 8.8 for every degree that the temperature increases in this model.



For my Multiple Linear Regression, I found that the most suited factors for predicting the natural log of Electrical Power were air temperature, house temperature, and total sun duration. This makes sense as these would be the most expected factors in how much Electrical Power would be used.



The values I found in these models are in my R markdown file provided.

Conclusions and Recommendations

After completing this project, I found that the best way of predicting the Electrical Power was through creating a linear model with the variables of were air temperature, house temperature, and total sun duration as these returned the greatest R-squared value of 0.1935 meaning that it represents 19.35% of the variance in energy consumption is explained by the model. I believe that I got a better understanding of how the different databases can be used for specific problems as well as being able to combine csv files and cleaning them to perform an analysis.

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